

Practical 2: Hardy Weinberg Equilibrium

Bioinformatics and Statistical Genetics (BSG)

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```
install.packages("HardyWeinberg")
```

```
## Loading required package: mice
```

```
##
```

```
## Attaching package: 'mice'
```

```
## The following object is masked from 'package:stats':
```

```
##
```

```
##      filter
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      cbind, rbind
```

```
## Loading required package: nnet
```

```
## Loading required package: Rsolnp
```

```
## Loading required package: shape
```

```
## Loading required package: combinat
```

```
##
```

```
## Attaching package: 'combinat'
```

```
## The following object is masked from 'package:utils':
```

```
##
```

```
##      combn
```

```
## Loading required package: gdata
```

```
##
```

```
## Attaching package: 'gdata'
```

```
## The following objects are masked from 'package:data.table':
```

```
##
```

```
##      first, last
```

```
## The following object is masked from 'package:stats':
##
##      nobs

## The following object is masked from 'package:utils':
##
##      object.size

## The following object is masked from 'package:base':
##
##      startsWith

## Loading required package: gtools

## Loading required package: MASS

## Loading required package: mvtnorm

##

## NOTE: THIS PACKAGE IS NOW OBSOLETE.

##

## The R-Genetics project has developed an set of enhanced genetics
## packages to replace 'genetics'. Please visit the project homepage
## at http://rgenetics.org for informtion.

##

##
## Attaching package: 'genetics'

## The following objects are masked from 'package:base':
##
##      %in%, as.factor, order
```

Loading dataset

```
df <- fread("TSIChr22v4.raw")
geno_dt <- as.data.table(df[, 7:ncol(df)])
```

1. How many variants are there in this database? What percentage of the data is missing?

```

n_variants <- ncol(geno_dt)

cat("Number of variants:", n_variants, "\n")

## Number of variants: 1102156

pct_missing <- mean(is.na(geno_dt)) * 100
cat("Percentage missing:", round(pct_missing, 4), "%\n")

## Percentage missing: 0 %

```

2. Calculate the percentage of monomorphic variants. Exclude all monomorphics from the database for all posterior computations of the practical. How many variants do remain in your database?

```

# Convert to matrix (faster than other options)
m <- as.matrix(geno_dt)

# For each column, count non-NA values equal to 0, 1, 2
has0 <- colSums(m == 0, na.rm = TRUE) > 0
has1 <- colSums(m == 1, na.rm = TRUE) > 0
has2 <- colSums(m == 2, na.rm = TRUE) > 0

# Monomorphic = only one of the three values appears
n_present <- has0 + has1 + has2
monomorphic <- n_present <= 1

cat("Monomorphic %:", mean(monomorphic) * 100, "\n")

## Monomorphic %: 81.03045

geno_poly <- geno_dt[, !monomorphic, with = FALSE]

n_poly <- ncol(geno_poly)
cat("Remaining variants:", n_poly, "\n", "Removed variants:", ncol(geno_dt)-n_poly, "\n", "These polymor")

## Remaining variants: 209074
## Removed variants: 893082
## These polymorphic variants were used for all subsequent analyses.

```

3. Extract polymorphism rs587756191 T from the data, and determine its genotype counts. Apply a chi-square test for Hardy-Weinberg equilibrium, with and without continuity correction. Also try an exact test, and a permutation test. You can use three functions HWChisq, HWExact and HWPPerm for this purpose. Do you think this variant is in equilibrium? Argue your answer

```
"rs587756191_T" %in% colnames(geno_poly)
```

```
## [1] TRUE
```

```
x <- geno_poly[["rs587756191_T"]]
```

```
# Genotype counts
```

```
g0 <- sum(x == 0, na.rm = TRUE)
```

```
g1 <- sum(x == 1, na.rm = TRUE)
```

```
g2 <- sum(x == 2, na.rm = TRUE)
```

```
counts <- c(AA = g0, AB = g1, BB = g2)
```

```
print(counts)
```

```
##   AA   AB   BB
```

```
## 106    1    0
```

```
# Chi-Square Test
```

```
# Without continuity correction
```

```
HWChisq(counts, cc = 0)
```

```
## Warning in HWChisq(counts, cc = 0): Expected counts below 5: chi-square
```

```
## approximation may be incorrect
```

```
## Chi-square test for Hardy-Weinberg equilibrium (autosomal)
```

```
## Chi2 =    0.0024 DF =    1 p-value =  0.9613
```

```
## f = -0.0047 Ho =  0.0093 He =  0.0093
```

```
# With continuity correction
```

```
HWChisq(counts, cc = 0.5)
```

```
## Warning in HWChisq(counts, cc = 0.5): Expected counts below 5: chi-square
```

```
## approximation may be incorrect
```

```
## Chi-square test with continuity correction for Hardy-Weinberg equilibrium (autosomal)
```

```
## Chi2 = 106.2512 DF =    1 p-value =  0.0000
```

```
## f = -0.0047 Ho =  0.0093 He =  0.0093
```

```
# Exact Test
```

```
HWExact(counts)
```

```
## Haldane Exact test for Hardy-Weinberg equilibrium (autosomal)
```

```
## using standard (SELOME) p-value
```

```
## sample counts: nAA = 106 nAB = 1 nBB = 0
```

```
## H0: HWE (D=0), H1: D <> 0
```

```
## D = 0.002336449 p-value = 1
```

```
# Permutation Test
HWPPerm(counts)
```

```
## Permutation test for Hardy-Weinberg equilibrium
## Observed statistic: 0.002358439    17000 permutations. p-value: 1
```

The variant rs587756191_T is in Hardy-Weinberg equilibrium.

4. Determine the genotype counts for all polymorphic variants, and store them in a $p \times 3$ matrix.

```
m <- as.matrix(geno_poly)

# Count genotypes per variant
n_AA <- colSums(m == 0, na.rm = TRUE)
n_AB <- colSums(m == 1, na.rm = TRUE)
n_BB <- colSums(m == 2, na.rm = TRUE)

# p x 3 matrix
genotype_counts <- cbind(AA = n_AA, AB = n_AB, BB = n_BB)

# Check dimensions
dim(genotype_counts)
```

```
## [1] 209074      3
```

```
# First few rows
head(genotype_counts)
```

```
##           AA AB BB
## rs587720402_A 106  1  0
## rs139377059_T 106  1  0
## rs587756191_T 106  1  0
## rs587702478_C 106  1  0
## rs62224609_C   91 16  0
## rs62224611_C   94 13  0
```

5. Apply an exact test for Hardy-Weinberg equilibrium to each SNP. You can use function HWExactStats for fast computation. What is the percentage of significant SNPs (use $\alpha = 0.05$)? Is this the number of markers that you would expect to be out of equilibrium by the effect of chance alone?

```
#Exact test for all SNPs
hwe_results <- HWExactStats(genotype_counts)

# Significant at alpha = 0.05
significant <- hwe_results < 0.05
```

```
n_significant <- sum(significant, na.rm = TRUE)
pct_significant <- mean(significant, na.rm = TRUE) * 100

cat("Number of significant SNPs:", n_significant, "\n")
```

```
## Number of significant SNPs: 5793
```

```
cat("Percentage of significant SNPs:", round(pct_significant, 2), "%\n")
```

```
## Percentage of significant SNPs: 2.77 %
```

```
cat("Expected by chance (5%):", round(0.05 * length(hwe_results), 0), "\n")
```

```
## Expected by chance (5%): 10454
```

Conclusion

The observed percentage (2.77%) is lower than the expected 5%, meaning there are fewer deviations from HWE than would be expected by chance. This suggests the data are relatively clean. ## 6. Which SNP is most significant according to the exact test results? Give its genotype counts. In which sense is this genotypic composition unusual?

```
# Find the minimum p-value
min_p <- min(hwe_results, na.rm = TRUE)
cat("Minimum p-value:", min_p, "\n")
```

```
## Minimum p-value: 9.784766e-33
```

```
# Which SNP has it
most_sig <- which.min(hwe_results)

cat("Most significant SNP:", colnames(geno_poly)[most_sig], "\n")
```

```
## Most significant SNP: rs2629366_C
```

```
genotype_counts[most_sig, ]
```

```
## AA AB BB
## 56 0 51
```

Most Significant SNP

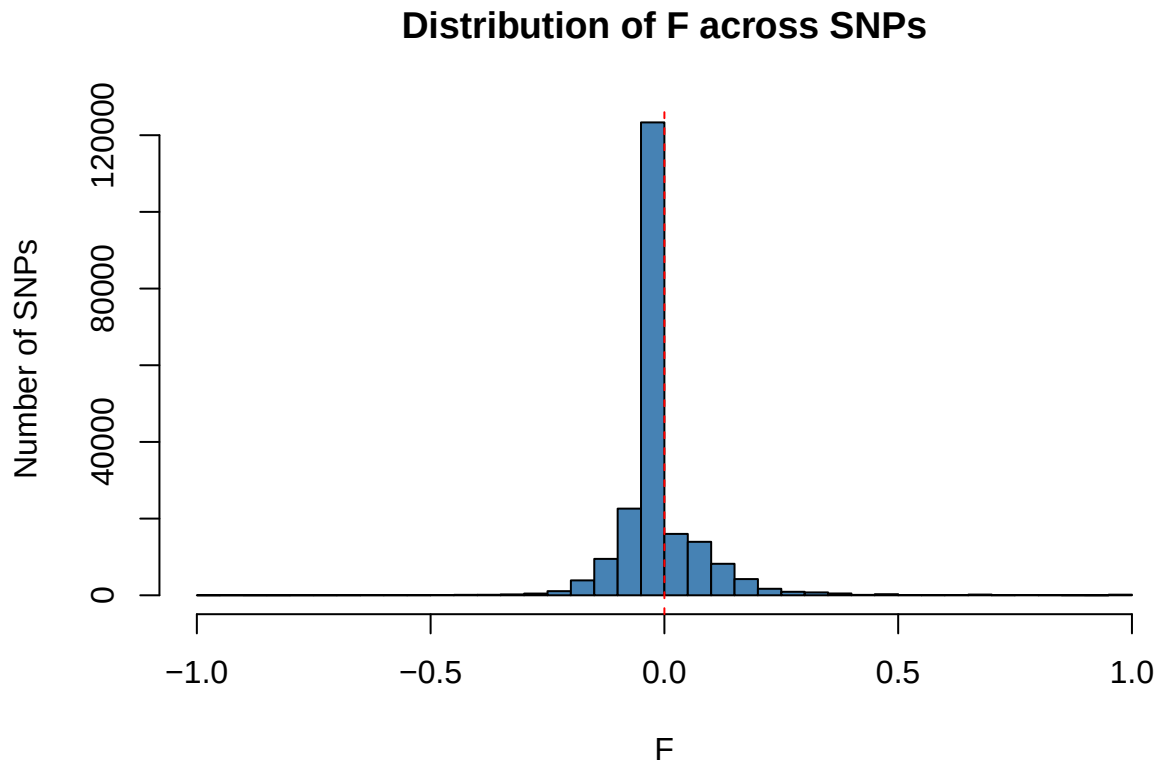
The most significant SNP according to the exact test has genotype counts of 56 AA, 0 AB, and 51 BB.

A complete absence of heterozygotes is biologically implausible under random mating and is almost certainly an artifact

7. Compute the inbreeding coefficient (f) for each SNP, and make a histogram of f . You can use function `HWf` for this purpose. Give descriptive statistics (mean, standard deviation, etc) of f calculated over the set of SNPs. What distribution do you expect f to follow theoretically? Use a probability plot to confirm your idea.

```
# F for all SNPs
f_values <- HWf(genotype_counts)

# Hist
hist(f_values, breaks = 50, col = "steelblue",
     main = "Distribution of F across SNPs",
     xlab = "F", ylab = "Number of SNPs")
abline(v = 0, col = "red", lty = 2)
```



```
# Descriptive statistics
cat("Mean:", round(mean(f_values, na.rm = TRUE), 4), "\n")
```

```
## Mean: -0.0047
```

```
cat("SD:", round(sd(f_values, na.rm = TRUE), 4), "\n")
```

```
## SD: 0.095
```

```
cat("Median:", round(median(f_values, na.rm = TRUE), 4), "\n")
```

```
## Median: -0.0047
```

```
cat("Min:", round(min(f_values, na.rm = TRUE), 4), "\n")
```

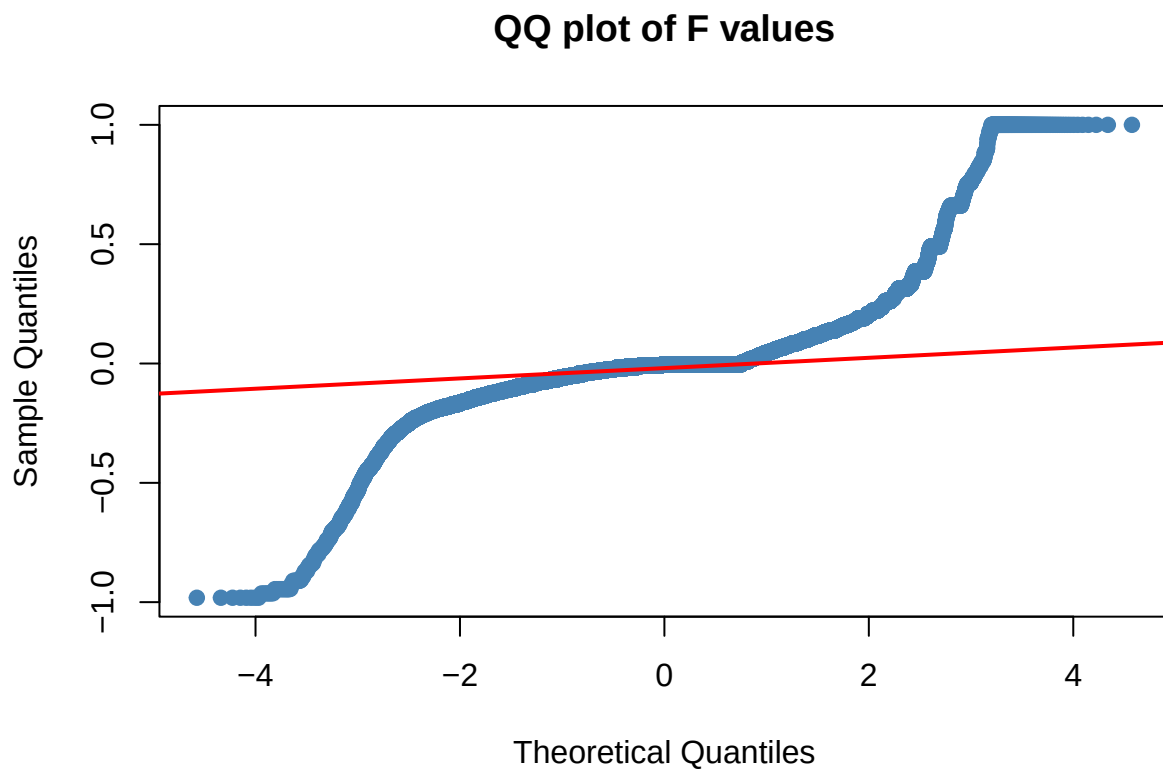
```
## Min: -0.9815
```

```
cat("Max:", round(max(f_values, na.rm = TRUE), 4), "\n")
```

```
## Max: 1
```

```
# QQ plot
```

```
qqnorm(f_values, main = "QQ plot of F values", pch = 19, col = "steelblue")  
qqline(f_values, col = "red", lwd = 2)
```



Some SNPs show extreme F values, suggesting real deviations from HWE

8. Apply the exact test for HWE to each SNP, using different significant levels. Report the number and percentage of significant variants using an exact test for HWE with $\alpha = 0.10, 0.05, 0.01$ and 0.001 . State your conclusions.


```

# exact test to all SNPs
hwe_results <- HWExactStats(genotype_counts)

# significance levels
alphas <- c(0.10, 0.05, 0.01, 0.001)

# significant SNPs at each level
results <- data.frame(
  alpha = alphas,
  n_significant = sapply(alphas, function(a) sum(hwe_results < a, na.rm = TRUE)),
  pct_significant = sapply(alphas, function(a) mean(hwe_results < a, na.rm = TRUE) * 100)
)

# expected by chance
results$expected_chance <- results$alpha * length(hwe_results)

print(results)

```

```

##   alpha n_significant pct_significant expected_chance
## 1 0.100          10049         4.8064322         20907.400
## 2 0.050           5793         2.7707893         10453.700
## 3 0.010           2508         1.1995753          2090.740
## 4 0.001           1485         0.7102748           209.074

```